

Is it really vital to get 8 hours of sleep a night? The largest analysis of brain scanning data yet casts doubt on the idea that shorter sleep duration is linked to shrinkage of the brain, reports **Clare Wilson**

EVERYONE knows that going to bed too late can leave you less alert the next day. But recent claims have gone further, promoting the idea that regularly getting too little sleep raises the risk of Alzheimer's disease and even shrinks the brain.

Now, this idea has been challenged by one of the most comprehensive analyses of brain scanning studies to date. So how much do we really need to worry about getting enough shut-eye?

Most health bodies, such as those in the US and the UK, advise that adults generally need between 7 and 9 hours of sleep a night. And, thanks to wrist-worn sleep tracking devices, it has never been easier to know if you are meeting this quota.

It is understandable that some people obsess over getting enough sleep. Widely covered animal research shows that the brain's cleaning system ramps up at night, ridding it of toxic compounds linked to Alzheimer's disease, such as a protein called beta-amyloid. And large population studies have generally found that people with unusually short or long sleep durations have worse health by several measures, including memory loss and brain shrinkage, one of the hallmarks of Alzheimer's.

Typically, there is a roughly U-shaped curve for the relationship between hours of slumber and the chances of a bad health outcome, such as dementia, and an inverted U-shaped curve for measures where a higher value signifies better health, such as brain volume seen on an MRI scan.

But such population studies can only find correlations between sleep duration and health. They can't tell us if poor sleep is causing the health condition. Only a randomised trial could do that, but these are almost impossible to set up in this case, because few people would agree to change their sleeping habits for



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any length of time in the name of science, says Anders Fjell at the University of Oslo in Norway.

Fjell and his team have now tried to take a deeper look through a series of studies that used brain volume as a proxy for brain health (*Nature Human Behaviour*, doi.org/k3df). First, they investigated brain volume in relation to sleep duration at a single point in time, using existing data on about 47,000 people. Here, they found an inverted U-shaped curve, although the

6.5 hours

Sleep duration linked to maximum brain volume in a new analysis

highest brain volume was linked with a surprisingly low 6.5 hours of sleep per night.

The team then carried out a further analysis on data that tracked about 4,000 people for up to 11 years. In this case, there was no correlation between sleep duration at the beginning of the study and brain shrinkage over this period. "It would be very surprising to see this, if short sleep had a negative effect on the brain," says Fjell.

The correlation seen in the first

When we sleep, the brain clears out waste substances

analysis could be explained by brain shrinkage causing sleep disturbance, rather than the other way around. Alternatively, it could just reflect stable differences between people, rather than the results of brain shrinkage, such as that people with naturally smaller or larger brains tend to sleep less for some unknown reason, he says.

The researchers also did a third kind of analysis of about 30,000 people, using genetic data gathered from the same sources as their first study. This found that those who are genetically predisposed to have either short or long sleep durations don't have smaller brain volumes than more moderate sleepers. Put together, this set of results challenges the idea that failing to get enough sleep shrinks the brain, says Fjell.

While he doesn't advise anyone to deliberately change their sleeping habits as a result of the findings, Fjell believes that there is much natural variation in people's innate sleep requirements and – unless prevented by circumstances – a person's brain will make sure it

gets the necessary amount.

"If you are feeling tired all day, then you probably sleep too little, and that can have negative consequences," he says. "But as long as you feel fine during the day, I wouldn't worry about whether your sleep is 6, 7 or 8 hours."

It is unlikely that the team's conclusions will convince everyone. Matthew Walker at the University of California, Berkeley, who wrote a book called *Why We Sleep* – which advises people to sleep for 8 hours a night – says the key measure of brain health is neuron density, rather than total brain volume. "Those two measures are not perfectly correlated with each other, which tells us that they measure different things," he says.

Walker also says the crucial measure of sleep quality is how long people spend in deep sleep – when brainwaves become slower – rather than their total quantity of sleep.

Certainly, it is during this slow-wave sleep that amyloid clearance from the brain intensifies. However, most such sleep happens during the first 4 hours of the night, says Fjell. "It is unclear whether more sleep than that would further facilitate this clearance," he says.

Maiken Nedergaard at the University of Rochester in New York says this is an important piece of work, due to its finding that, in the single-time-point analysis, larger brain volume was linked with a mere 6.5 hours of sleep a night. "The surprise is that the optimal sleep duration is shorter than what is currently recommended," she says.

This result suggests, therefore, that even if you are concerned about total sleep duration, you may not need to worry about falling short of 8 hours a night. "You should not lie in bed tossing and turning to meet [that] target," says Michael Chee at the National University of Singapore. "This study is quite reassuring." ■